→ Non-linear svm

data points can't seperate with simple straight line. Insuch cases svm uses kernel trick.

Kernel trick – it is kernel function to transform the data into a Higher – dimensional feature space. Where a linear seperator can be found.

Original space → kernel transformation → Higher dimensional space → Linear seperation

Kernel trick → helps find a linear boundary in a transformed feature space.

Eg: Original features are

x_1, x_2

these data is not linearly separable

$$x_3 = x_1^2 + x_2^2$$

Now we have $(x_1, x_2) \rightarrow (x_1, x_2, x_3)$

2D → 3D

Why do we need the kernel trick

We explicitly transform every data point into a very Higher dimensional space, that could be computationally expensive.

feature space

Instead of doing $x \rightarrow \phi(x)$

then calculating $\phi(x_i)^T \phi(x_j)$

we directly calculating.

$K(x_i, x_j)$

Where $K(x_i, x_j) = \phi(x_i)^T \phi(x_j)$

Kernel trick directly calculates Similarity between data points.

Important Kernels:-

Linear kernel $\Rightarrow$ $K(x_i, x_j) = x_i^T x_j$

used when data is linearly separable.

(2) Polynomial kernel

$$K(x_i, x_j) = (x_i^T x_j + c)^d$$

d = degree, γ = constant

It can model more complex boundaries

(3) RBF / Gaussian kernel

· Commonly used

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

points that are close together have high similarity

point that are apart have low similarity

(4) sigmoid kernel

$$K(x_i, x_j) = \tanh(\gamma x_i^T x_j + c)$$

Feature space → It is mathematical space in which we represent our data using its features

2 features → 2D features space

3 feature → 3D feature space.

linear SVM boundary

$$w^T x + b = 0$$

$w^T x + b > 0 \rightarrow +1$ + positive class

$w^T x + b < 0 \rightarrow -1$ − ve class

→ ⑤ Hard Margin linear SVM,

Optimization is

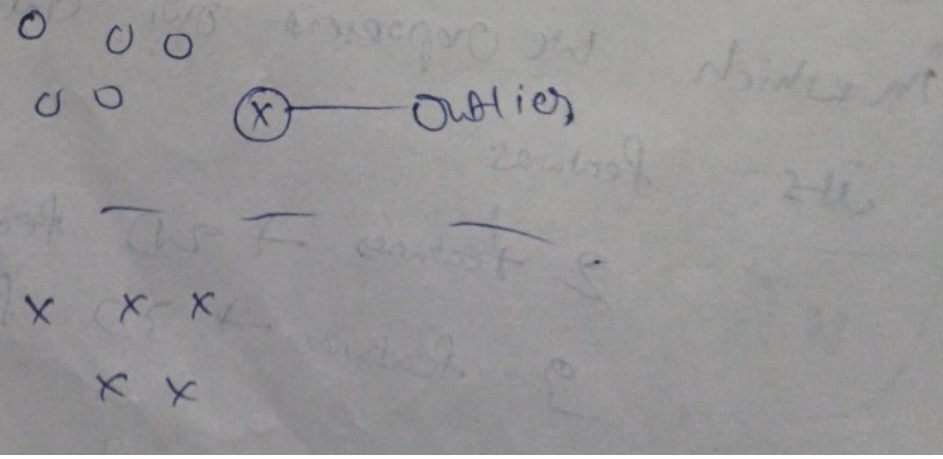
$$\min \frac{1}{2} \|w\|^2$$

Subject to

$$y_i (w^T x + b) \geq 1$$

every training point must be classified correctly

Case what if there is an outlier



Soft margin SVM

It allows some points to violate the margin & even be misclassified.

This introduces slack variable ξ_i

And the objective function becomes

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i$$

'C' parameter controls the trade off between a wider margin and penalties for violations.

In Hard margin SVM.

$$y_i(w^T x_i + b) \geq 1$$

It means every training point must be correctly classified and outside the margin

In real world data may contain
→ noise, outliers, overlapping classes.
Perfect seperation may not be possible

SVM - allows some violations using slack variable ξ_i

ξ tells us how much a particular training point violates the margin requirement

Soft margin constraint becomes.

$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

Case 1: $\xi_i = 0$

$$y_i(w^T x_i + b) \geq 1$$

The point satisfies the margin

No violation.

Case 2: $0 < \xi_i < 1$

The point correctly classified but inside the margin.

$\xi > 1$ → Margin violation.

Case 3:- The point crossed the decision boundary.

It is misclassified

Objective

$$\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\varepsilon$$

- $\frac{1}{2}\|w\|^2$ — 1st part — Large — Controls margin
- $C\sum_{i=1}^{n}\varepsilon$ — penalty for violations.

$$\text{Margin} = \frac{2}{\|w\|}$$ — minimize

2nd part — penalty margin violations

$$C\sum_{i}\varepsilon_i$$

C is penality / importance given to mistakes

$C \rightarrow$ large.

fewer training errors, potentially a narrower margin.

Small C

$C \rightarrow$ small

Wider margin, with potentially more training violations.

Eg Model A

Wide margin but 3 points violate the margin.

$\frac{1}{2}\|w\|^2 = 2$

$\sum \xi_i = 3$

$C = 1$

Total objective $= 2 + 1(3) = 5$

Model - B

Narrower margin but no violations

$$\frac{1}{2}\|w\|^2 = 6$$

and $\sum \xi_i = 0$

Objective

$$6 + (0) = 6$$

SVM chooses smaller objective,

Model 'A' preferred in this

Increase $C = 10$

for model A

$$2 + 10(3) = 32$$

for model B

$$6 + 10(0) = 6$$

Now model is preferred

Small C — Can more about keeping wide margin. more tolerance for mistakes

Large C — more about avoiding violations. less tolerance for mistakes

Kernel

A kernel function calculates the similarity between two data points.

$K(x_i, x_j)$ — first point, second point

\ Kernel function

O/P — Similarity value b/w data points

Kernel trick — calculates similarity in a transformed feature space. without explicitly creating the tranformed features

Linear kernel :-

$$K(x_i, x_j) = x_i^T x_j$$

$x_i = [2, 3]$ $x_j = [4, 5]$

$x_i^T x_j = (2 \times 4 + 3 \times 5) = (8 + 15)$

$K(x_i, x_j) = 23$ — Similarity value

Polynomial kernel :-

$$K(x_i, x_j) = (x_i^T x_j + c)^d$$

$x_i^T x_j$ = dot product

c = constant

d = degree of polynomial

$x_i = [2, 3]$, $x_j = [4, 5]$

$d = 2$, $c = 1$

$x_i^T x_j = (2 \times 4) + (3 \times 5)$

$= 23$

$K(x_i, x_j) = (23 + 1)^2$

$= (24)^2 \Rightarrow 576$

$d = 3$

$$K(x_i, x_j) = (x_i^T x_j + c)^3$$

Higher degree allows SVM to model more complex relationships, but too high a degree can make the model unnecessarily complex.

RBF kernel:-

Radial Basis kernel also called

Gaussian kernel

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

$x_i - x_j \Rightarrow$ finds the difference between two points

$\|x_i - x_j\|^2 \Rightarrow$ represent squared Euclidean distance between two points

$-\gamma \|x_i - x_j\|^2 \Rightarrow$ controls how quickly similarity decreases as distance increases

Eg $x_i = [1, 2]$ $x_j = [2, 4]$

$\gamma = 0.5$

$x_i - x_j = [1-2, 2-4] = (-1, -2)$

Step-2 Square and add. $1 + 4 = 5$

Step-3 multiply by $\gamma = -0.5(5) = -2.5$

Step-4 apply exponential

$K(x_i, x_j) = e^{-2.5}$

$K(x_i, x_j) \approx 0.082$

So the similarity relatively small because the two points are relatively far apart

& Suppose two points are very close.

$$\|x_i - x_j\| \approx 0$$

$$e^{-\gamma \|x_i - x_j\|^2}$$

$$e^{-0} = 1$$

$$k(x_i, x_j) = 1$$

very close points → similarity close to 1

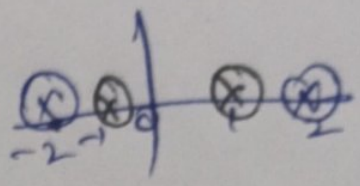
very far away → similarity close to 0

Small γ — decision boundary tends to be smooth

Large γ — tightly shaped decision boundary

Eg

x	class
−2	+ − red
+2	+ − red
−1	− − blue
+1	− − blue



$z = x^2$ — new Dimension

x	class	x^2	
−2	+	4	+ class
+2	+	4	
−1	−	1	− class
+1	−	1	

$z = 2.5$ — boundary sperate Classes